

Unit 9: Igneous Rocks

Reading: pages 125–133 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. The rock cycle is driven by two forces. They are
 - a) Earth's internal heat engine and the hydrological cycle
 - b) plate tectonics and erosion
 - c) the Sun and the Moon
 - d) gravity and magnetism
2. Why is the rock cycle virtually dead on the Moon?
 - a) The Moon is too small to have ever had rocks
 - b) Its surface is entirely covered by ice
 - c) The Moon contains no minerals
 - d) Its core is no longer hot enough to drive mantle convection, and it has no atmosphere or liquid water
3. Igneous rock that cools slowly deep beneath Earth's surface is described as
 - a) intrusive or plutonic
 - b) aphanitic
 - c) extrusive
 - d) volcanic
4. An irregularly shaped pluton with an exposed surface area greater than 100 km² is called
 - a) a stock
 - b) a batholith
 - c) a sill
 - d) a dyke
5. A tabular, sheet-like pluton that is *concordant* with existing layering in the country rock is
 - a) a dyke
 - b) a sill
 - c) a xenolith
 - d) a batholith

6. Which extrusive igneous rock contains holes left by trapped gas bubbles, making it light enough to float on water?
- a) Obsidian
 - b) Basalt
 - c) Pumice
 - d) Granite

Part B · True or False

2 questions × 1 point = 2 points

7. In general, the longer a body of magma takes to cool, the larger its crystals will be.
TRUE / FALSE
8. Basalt is the most common intrusive igneous rock.
TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Explain what the textures *phaneritic*, *aphanitic*, and *porphyritic* each tell you about how a rock cooled.
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10. A body of mafic magma cools slowly underground. Which minerals form, and what rock results? What rock forms instead if that same magma cools quickly at the surface?
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